7COM1033: Neural Networks and Machine Learning
Formative Assessment on Essential Mathematics, Semester B 2025
Time: 90 minutes
Total marks: 77(2 bonus); 4 questions; 2 pages of questions
Show all working on paper. There are easy parts and trickier parts all mixed up.
If you are stuck, skip and then return to harder parts later.

1. Vectors and Matrices
   (a) Given the following matrices:
   $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$
   $B = \begin{pmatrix} 5 & 0 & 1 \\ 3 & -2 & 0 \end{pmatrix}$
      i. (3 points) Find AB.
      ii. (1 point) Is it possible to fine BA? If not, why not?
   (b) (1 point) Given the following matrix, what is $W_{2,3}$?
   $$W = \begin{pmatrix} 2 & 2 & -3 & 0 \\ -1 & -1 & -1 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -\frac{1}{5} & 5 \\ 1 & -1 & -3 & 0 \end{pmatrix}$$
   (c) (2 points) Which elements of W are equal to -3?(Be sure to use the correct notation).
   (d) (3 points) What is the dot product of the last two columns of W? What is the angle between these vectors (does this relationship between vectors have a special name)?
   (e) (5 points) Compute the angle between the vectors comprising the 2nd and 3rd rows of W (Hint: take the dot product).
   (f) (3 points) Compute the vector $y_a = \sum_{b=1}^{4} W_{ab}x_b$, where and $x = (1, -1, 0, 1)^T$. How many coordinates does it have? (Is it a row vector or a column vector?)

2. Lines and hyperplanes
   (a) (2 points) What is the y-intercept and gradient of the line $y = 5x - 17$?
   (b) (2 points) What is the equation of the line in $\mathbb{R}^2$ that goes through the points $(0, 3)$ and $(4, 0)$?
   (c) (2 points) What is the equation of the line in $\mathbb{R}^2$ that goes through the points (x_1, y_1) and (x_2, y_2)?
   (d) (2 points) Provide a normal vector to the line $y = 5x - 17$ when considered a hyperplane in $\mathbb{R}^2$. Is it unique?
   (e) (2 points) What is the normal vector of the hyperplane given by the same equation, $y = 5x - 17$, but in $\mathbb{R}^3$?
   (f) (2 points) BONUS: Write down two linearly independent vectors that lie on the plane $y = 5x - 17$ in $\mathbb{R}^3$.
   (g) (2 points) What is the normal vector to the hyperplane in $\mathbb{R}^5$ given by the equation:
   $$2x_1 + 4x_2 - x_3 + 7x_4 + x_5 - 10 = 0?$$

(h) (2 points) Which of the following points lies on this hyperplane $E = (-1,-1,-10,1,-1)$ or $F = (0,0,0,0,0)$? Explain why.

(i) (3 points) In $\mathbb{R}^3$ what is the distance between the points $C = (2,4,9)$ and $D(0,1,3)$?

(j) (4 points) In $\mathbb{R}^n$ (n-dimensional real Euclidean space), a line can be defined by a parametric equation of the form:

$$\vec{x} = \lambda\vec{v} + \vec{d},$$

where $\lambda \in \mathbb{R}$. Give the equation of the line going through points C and D as defined above.

3. Differentiation

(a) (2 points) Differentiate $y = 2x^7$.

(b) (2 points) Differentiate $y = (5x+1)^3 + \exp(x)$

(c) (2 points) Differentiate $y = \frac{1}{x} - 2\ln(x)$.

(d) (2 points) Differentiate $y = \sqrt{(2x^2 - 5x + 8)}$

(e) (2 points) If $g(x) = \sin(x)$ and $h(x) = 4x$ for $x \in \mathbb{R}$, what is $g \circ h(x)$ and $h \circ g(x)$?

(f) (4 points) Carefully state the chain rule and find the derivatives, $(g \circ h)'(x)$ and $(h \circ g)'(x)$.

(g) (4 points) For $f(x) = \frac{x}{10}$, what is $f \circ f \circ f(x)$? Is it the same as $[f(x)]^3$? Explain your answer.

(h) (4 points) Find the derivatives of $f \circ f \circ f(x)$ and $[f(x)]^3$.

4. Partial derivatives

(a) (4 points) Given the following function $f : \mathbb{R}^3 \to \mathbb{R}$,

$$f(x,y,z) = xyz + 5\sin(xy) + \ln(3z)$$

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compute ∇f (the vector of partial derivatives).

(b) (4 points) In $\mathbb{R}^2$ the relationship between Cartesian coordinates (x,y) and polar coordinates (r,θ) is given via the following transformation:

$$x = r\cos\theta,\ y = r\sin\theta.$$

Compute all the partial derivatives $\frac{\partial x}{\partial r}$, $\frac{\partial y}{\partial r}$, $\frac{\partial x}{\partial \theta}$, and $\frac{\partial y}{\partial \theta}$ and write down the Jacobian $\frac{\partial(x,y)}{\partial(r,\theta)}$.

(c) (2 points) What is the determinant of the Jacobian?

(d) (4 points) Express r and θ in terms of x and y which is the inverse transformation going the other way.